QTherm-X: Explainable Quantum-Assisted Physics-Informed Neural Networks for Thermal Physics

A Harikiran, A Immanuel Prince, R Jeyaprasanna
Department of Artificial Intelligence and Data Science
Panimalar Engineering College
Chennai, India
harikiran1328@gmail.com
imman.augustin@gmail.com
okprasanna8@gmail.com

***Abstract*—Physics-Informed Neural Networks (PINNs) have emerged as an effective approach for solving partial differential equations by incorporating physical laws into neural network optimization. Recent developments in quantum computing have introduced opportunities to enhance scientific machine learning through hybrid quantum-classical architectures.**

This paper presents QTherm-X, an Explainable Quantum-Assisted Physics-Informed Neural Network (QAPINN) framework for thermal physics applications. The proposed framework integrates a Variational Quantum Circuit implemented using PennyLane with a classical PINN developed using PyTorch for solving the one-dimensional heat equation.

The hybrid architecture utilizes quantum feature extraction through parameterized quantum circuits while maintaining physical consistency using PDE-based loss functions. A benchmark comparison is performed between Classical PINN and Hybrid QAPINN models using different quantum circuit configurations.

The proposed framework is further investigated for aerospace thermal monitoring applications through NASA C-MAPSS dataset analysis. Experimental results demonstrate the feasibility of integrating quantum-assisted learning with physics-informed neural networks for thermal modelling and scientific machine learning applications.

***Index Terms*—Quantum Computing, Physics-Informed Neural Networks, Quantum Neural Networks, Thermal Physics, PennyLane, PyTorch, Scientific Machine Learning, NASA C-MAPSS**

I. Introduction

Thermal modelling plays a critical role in engineering applications including aerospace propulsion systems, marine engines, industrial manufacturing, electronic cooling, and energy systems. Accurate prediction of temperature distribution and heat transfer behaviour enables improved reliability, efficiency, and predictive maintenance.

Traditional numerical methods such as Finite Difference Methods (FDM) and Finite Element Methods (FEM) provide accurate solutions for thermal problems. However, these approaches often require high computational cost when dealing with complex geometries, nonlinear systems, and large-scale simulations.

Physics-Informed Neural Networks (PINNs) have emerged as a powerful alternative for solving physical problems by embedding governing equations directly into neural network training. Unlike conventional machine learning methods that require large labelled datasets, PINNs utilize physical constraints including differential equations, initial conditions, and boundary conditions.

PINNs have demonstrated significant success in solving forward and inverse problems involving fluid dynamics, heat transfer, elasticity, and other scientific applications.

Recent progress in quantum computing has introduced new possibilities for enhancing machine learning models. Variational Quantum Circuits (VQCs) provide trainable quantum transformations capable of representing complex nonlinear feature spaces. Hybrid quantum-classical neural networks combine quantum feature extraction with classical optimization techniques.

In this work, we propose QTherm-X, an Explainable Quantum-Assisted Physics-Informed Neural Network framework for thermal physics applications.

The proposed system combines:

- A Variational Quantum Circuit implemented using PennyLane.
- A Physics-Informed Neural Network implemented using PyTorch.
- Physics-based optimization using the one-dimensional heat equation.
- Benchmark comparison between Classical PINN and Hybrid QAPINN.

The major contributions of this work are:

- Development of an explainable Hybrid Quantum-Assisted PINN framework for thermal physics.
- Integration of quantum feature extraction with physics-informed learning.
- Benchmark evaluation of Classical PINN and Hybrid QAPINN using multiple quantum configurations.
- Investigation of aerospace thermal monitoring applications using NASA C-MAPSS datasets.

II. Background

A. Thermal Physics

Thermal physics describes the transfer and distribution of heat within physical systems.

Many engineering systems including aircraft engines, marine propulsion systems, and industrial equipment are governed by heat transfer equations.

The one-dimensional heat equation is commonly used as a fundamental model for analysing temperature evolution over space and time.

B. Physics-Informed Neural Networks

Physics-Informed Neural Networks integrate physical laws into deep learning models by incorporating governing equations into the loss function.

The general PINN framework consists of a neural network approximation:

$$u(x,t) = NN(x,t) \tag{1}$$

where the neural network learns the temperature field while satisfying the governing physical constraints.

The total loss function consists of:

$$\mathcal{L}_{total} = \mathcal{L}_{PDE} + \mathcal{L}_{IC} + \mathcal{L}_{BC} \tag{2}$$

where:

- $\mathcal{L}_{PDE}$ represents the physics equation residual.
- $\mathcal{L}_{IC}$ represents initial condition error.
- $\mathcal{L}_{BC}$ represents boundary condition error.

C. Quantum Neural Networks

Quantum Neural Networks utilize parameterized quantum circuits to perform nonlinear transformations in quantum feature space.

A Variational Quantum Circuit consists of trainable quantum gates whose parameters are optimized using classical optimization algorithms.

The hybrid quantum-classical approach combines quantum representation capability with classical neural network optimization.

D. Hybrid Quantum-Assisted PINNs

A Hybrid Quantum-Assisted Physics-Informed Neural Network combines a classical PINN with a quantum feature extraction layer.

The input variables are encoded into quantum states using quantum embedding techniques.

The resulting quantum measurements are converted into feature vectors and processed by classical neural network layers to predict the solution.

III. Related Work

PINNs have become an important research direction in scientific machine learning due to their ability to solve differential equations while maintaining physical consistency.

Previous studies have applied PINNs to problems involving heat transfer, fluid mechanics, structural mechanics, and inverse modelling.

Quantum machine learning has recently gained attention due to the development of near-term quantum devices and hybrid quantum-classical algorithms.

Variational Quantum Circuits have been explored for classification, regression, and scientific computing tasks.

Quantum Physics-Informed Neural Networks represent an emerging research area that combines quantum computing with physics-based machine learning.

However, existing approaches primarily focus on theoretical demonstrations using simple benchmark equations.

QTherm-X extends these concepts by investigating hybrid quantum-assisted PINNs for thermal physics modelling and aerospace-related thermal analysis.

IV. Proposed QTherm-X Framework

The proposed QTherm-X framework integrates Physics-Informed Neural Networks with a Variational Quantum Circuit to solve thermal physics problems while maintaining physical consistency.

The framework consists of three major components:

- Hybrid quantum-classical architecture.
- Physics-informed computational workflow.
- Variational quantum feature extraction module.

A. Overall System Architecture

The overall architecture of QTherm-X is illustrated in Fig. 1.

The input variables consisting of spatial coordinate (x) and temporal coordinate (t) are provided to the quantum feature extraction layer.

The Variational Quantum Circuit performs quantum state encoding using Angle Embedding and extracts quantum features through trainable quantum operations.

The generated quantum feature vector is then processed by classical neural network layers to approximate the temperature solution $u(x,t)$.

The complete framework is optimized using physics-informed loss functions derived from the governing heat equation.

B. Computational Workflow

The computational workflow of QTherm-X is shown in Fig. 2.

Initially, the thermal problem is defined using the governing partial differential equation.

The spatial and temporal coordinates are encoded into the hybrid quantum-classical model.

The quantum circuit generates feature representations which are passed to the classical neural network.

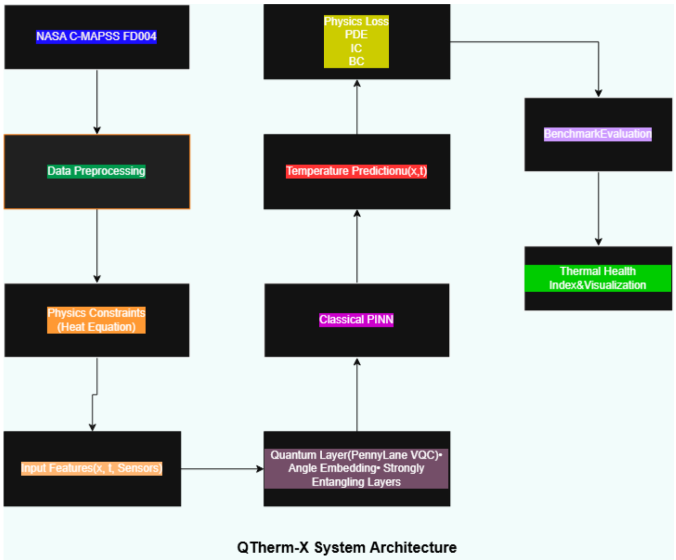

Fig. 1. Overall architecture of the proposed QTherm-X framework.

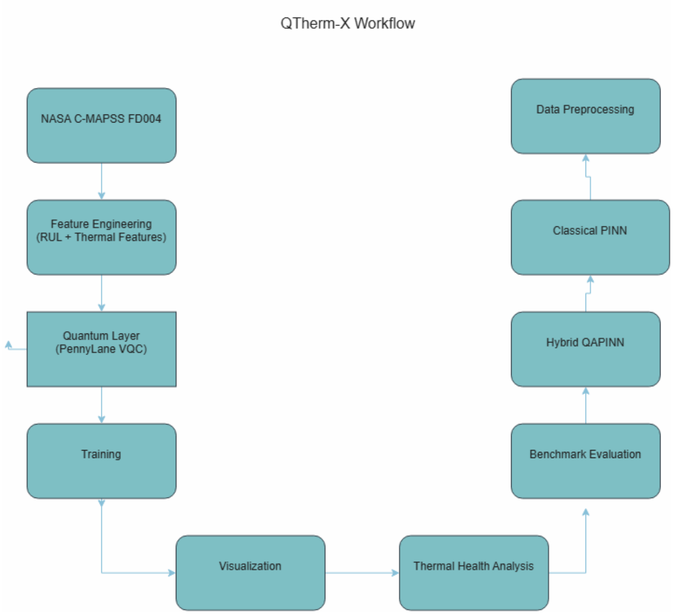

Fig. 2. Computational workflow of the proposed QTherm-X framework.

The predicted temperature field is optimized by minimizing the physics-informed loss consisting of PDE residual loss, initial condition loss, and boundary condition loss.

C. Variational Quantum Circuit

The quantum component of the Hybrid QAPINN is implemented using PennyLane.

The input vector (x, t) is encoded into quantum states using Angle Embedding.

The encoded quantum states are processed using Strongly Entangling Layers consisting of trainable rotation gates and entangling operations.

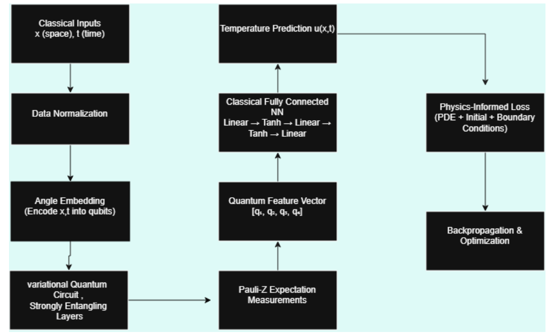

Fig. 3. Variational Quantum Circuit used in Hybrid QAPINN.

The quantum circuit output is obtained using Pauli-Z expectation measurements.

The resulting quantum feature vector is represented as:

$$Q = [\ \langle Z_1 \rangle, \langle Z_2 \rangle, ..., \langle Z_n \rangle]$$
(3)

These extracted features are provided to the classical neural network for temperature prediction.

V. Mathematical Formulation

The proposed Hybrid QAPINN solves the one-dimensional transient heat equation while satisfying physical constraints through physics-informed optimization.

A. Heat Equation

The governing thermal equation is defined as:

$$\partial u(x,t)_{\overline{\partial t = \alpha \frac{\partial^2 u(x,t)}{\partial x^2}}}$$
(4)

where:

- $u(x, t)$ represents temperature distribution.
- x represents spatial coordinate.
- t represents time.
- α represents thermal diffusivity.

B. Physics-Informed Loss Function

The total loss function consists of three components:

$$\mathrm{L}_{total} = \mathcal{L}_{PDE} + \mathcal{L}_{IC} + \mathcal{L}_{BC}$$
(5)

The PDE residual is calculated as:

$$r(x,t) = \partial u \frac{}{\partial t - \alpha \frac{\partial^2 u}{\partial x^2}} \quad (6)$$

The PDE loss is:

$$L_{PDE} = \frac{1}{N} \sum_{i=1}^{N} r(x_i, t_i)^2 \quad (7)$$

The initial condition loss is:

$$L_{IC} = \frac{1}{N} \sum_{i=1}^{N} (u(x_i, 0) - u_0(x_i))^2 \quad (8)$$

The boundary condition loss is:

$$L_{BC} = \frac{1}{N} \sum_{i=1}^{N} (u(0, t_i) - u(1, t_i))^2 \quad (9)$$

C. Quantum Feature Extraction

The quantum layer transforms the classical input variables into quantum feature representations.

For an n qubit system, the quantum circuit produces:

$$F_q = [f_1, f_2, ..., f_n] \quad (10)$$

where each feature corresponds to the expectation value measured from a quantum observable.

The quantum features are combined with classical neural network layers to predict the final temperature solution.

TABLE I
EXPERIMENTAL CONFIGURATION OF QTHERM-X

Parameter	Value
Quantum Framework	PennyLane
Classical Framework	PyTorch
Optimizer	Adam
Learning Rate	0.001
Training Epochs	3000
Training Points	5000
Quantum Layers	2
Qubit Configurations	2,3,4

D. Hybrid Model Training

The Hybrid QAPINN is trained using gradient-based optimization.

Automatic differentiation is used to compute the derivatives required for evaluating PDE residuals.

Both classical neural network parameters and quantum circuit parameters are optimized simultaneously using the Adam optimizer.

The training process continues until convergence while minimizing the physics-informed loss.

VI. EXPERIMENTAL SETUP

A. Development Environment

The proposed QTherm-X framework was implemented using Python.

The classical neural network components were developed using PyTorch, while the quantum circuit implementation was performed using PennyLane.

All experiments were executed using CPU-based computation.

B. Hybrid QAPINN Configuration

The Hybrid QAPINN architecture consists of a Variational Quantum Circuit followed by classical neural network layers.

The experimental configuration is summarized below:

C. Benchmark Models

Two models are evaluated in this work:

- Classical Physics-Informed Neural Network (PINN).
- Hybrid Quantum-Assisted Physics-Informed Neural Network (QAPINN).

Both models are trained using identical training points, optimization settings, and physics constraints to ensure a fair comparison.

VII. RESULTS AND DISCUSSION

A. Benchmark Comparison

The proposed Hybrid QAPINN was benchmarked against a Classical PINN model.

The experiments investigated the effect of different quantum circuit sizes using 2, 3, and 4 qubit configurations.

The final convergence results after 3000 training epochs are presented in Table II.

TABLE II
BENCHMARK COMPARISON BETWEEN CLASSICAL PINN AND HYBRID QAPINN

Model	Qubits	Final Loss
Classical PINN	-	6.733710×10^{-4}
Hybrid QAPINN	2	7.078764×10^{-2}
Hybrid QAPINN	3	9.090878×10^{-2}
Hybrid QAPINN	4	9.230375×10^{-2}

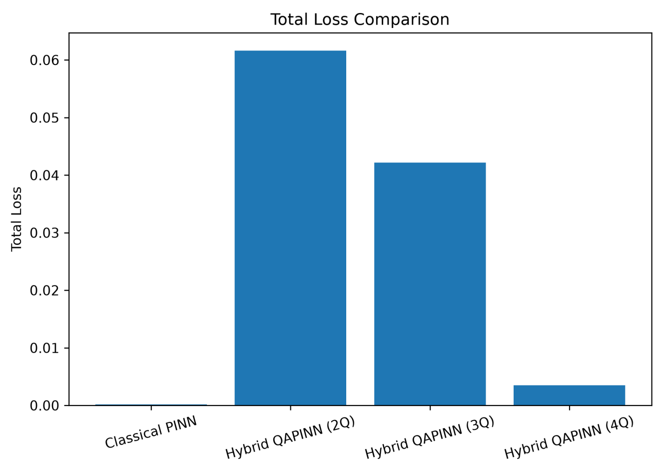

Fig. 4. Comparison of total training loss.

The Classical PINN achieved the lowest final optimization loss due to its simpler classical architecture.

The Hybrid QAPINN models demonstrated stable convergence while introducing quantum feature extraction through Variational Quantum Circuits.

These results demonstrate the feasibility of integrating quantum computing components with physics-informed neural networks for thermal PDE modelling.

B. Training Convergence Analysis

The convergence behaviour of the models was analysed using the training loss evolution.

The Classical PINN showed faster convergence, while the Hybrid QAPINN models maintained stable optimization behaviour throughout training.

The additional computational complexity introduced by quantum circuit evaluation affects convergence speed; however, the hybrid models successfully preserved the physics constraints.

C. Prediction Accuracy Evaluation

Prediction accuracy is evaluated using error-based measurements including Mean Squared Error (MSE), Mean Absolute Error (MAE), and Relative L_2 error.

These metrics quantify the difference between predicted temperature distributions and the expected physical solution.

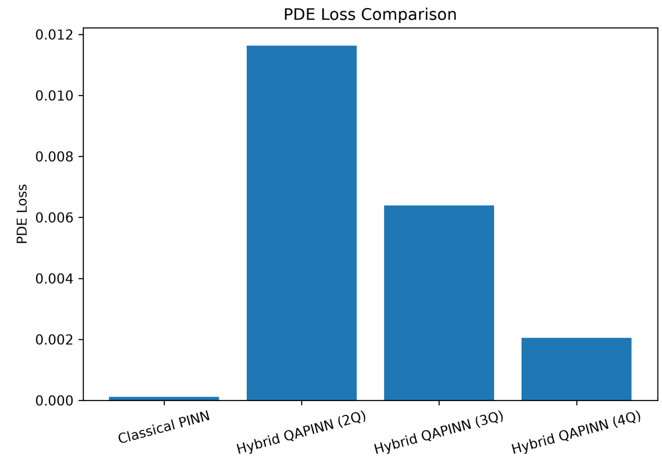

Fig. 5. Comparison of PDE residual loss.

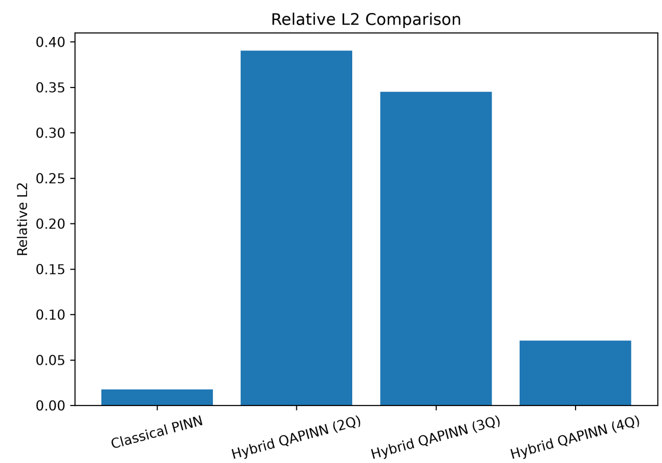

Fig. 6. Relative L_2 error comparison.

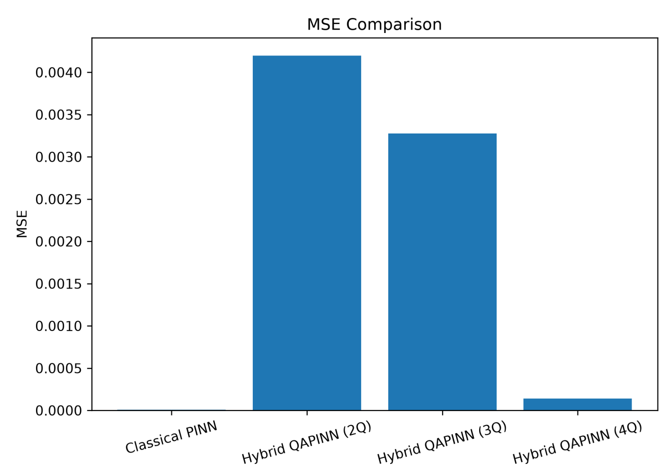

Fig. 7. Mean Squared Error comparison.

D. Computational Performance

Computational performance was analysed using training time and model complexity.

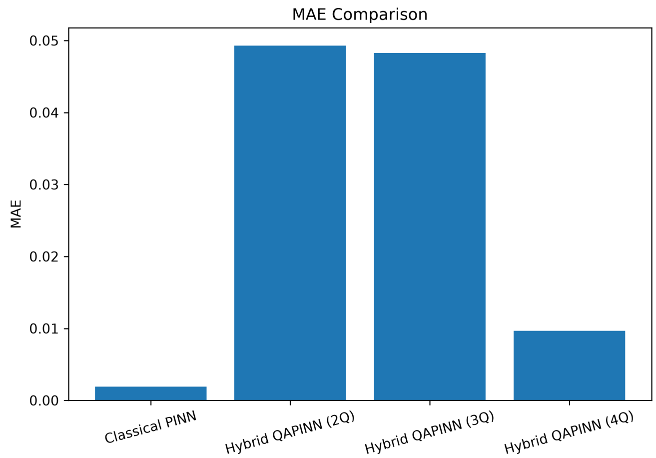

Fig. 8. Mean Absolute Error comparison.

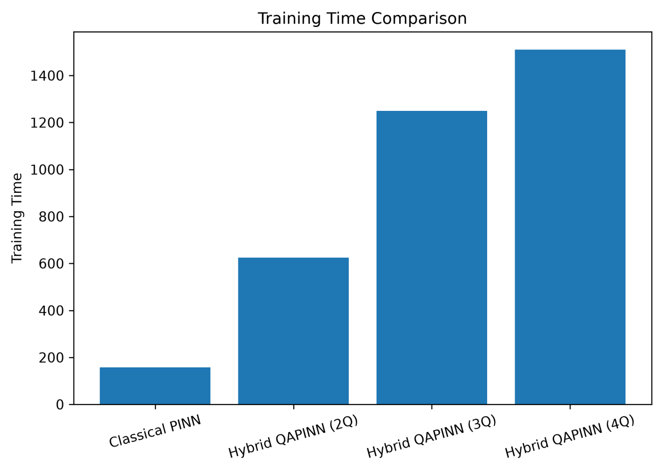

Fig. 9. Training time comparison between models.

The number of trainable parameters provides information about the computational requirements of each architecture.

Although Hybrid QAPINN introduces quantum circuit evaluation overhead, it provides an alternative feature extraction mechanism for scientific machine learning applications.

E. Thermal Field Visualization

The predicted thermal field generated by QTherm-X is visualized using two-dimensional heatmaps and three-dimensional surface plots.

These visualizations provide an intuitive understanding of temperature evolution across spatial and temporal dimensions.

VIII. NASA C-MAPSS Validation

A. Dataset Description

To investigate the applicability of QTherm-X for aerospace thermal monitoring scenarios, the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset is considered.

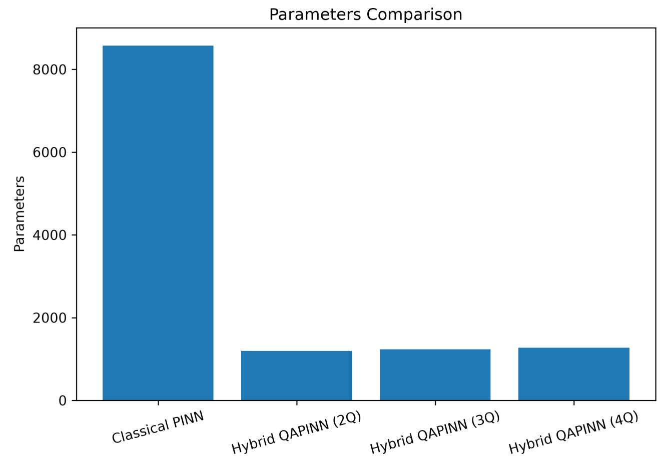

Fig. 10. Trainable parameter comparison.

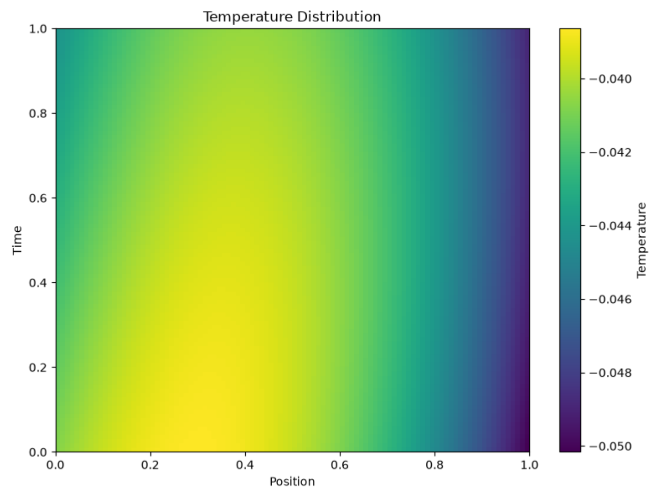

Fig. 11. Predicted temperature distribution heatmap.

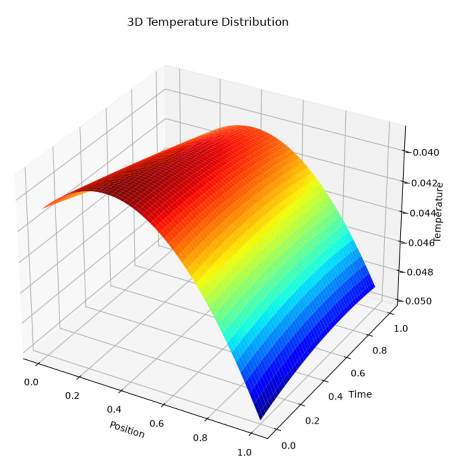

Fig. 12. Three-dimensional thermal surface visualization.

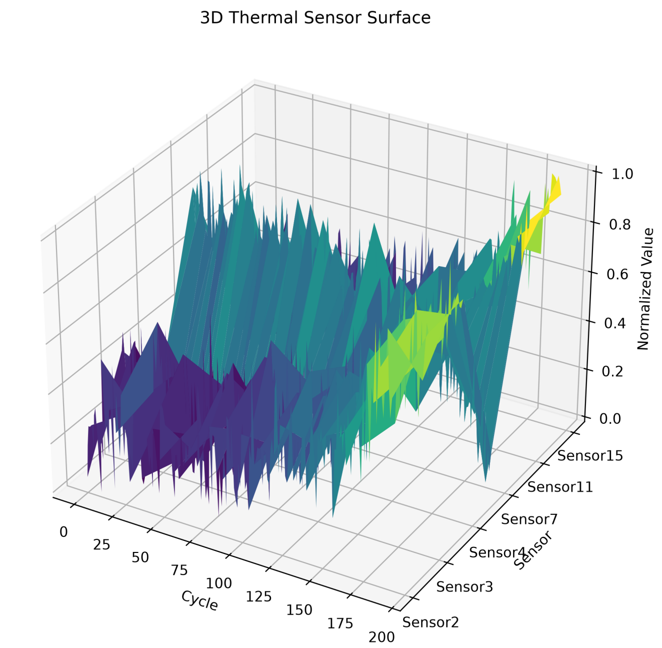

Fig. 13. Thermal surface visualization for NASA C-MAPSS FD001 dataset.

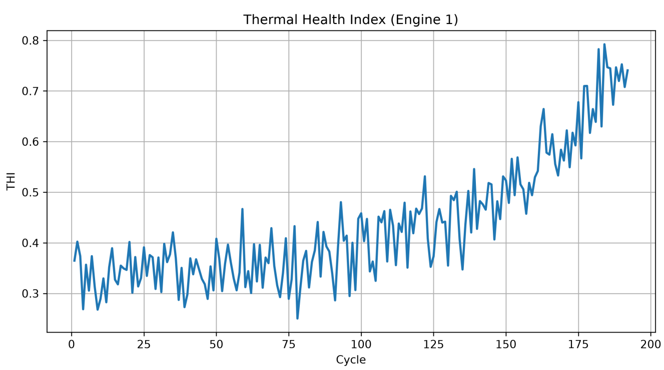

Fig. 14. Thermal health indicator analysis for FD001 dataset.

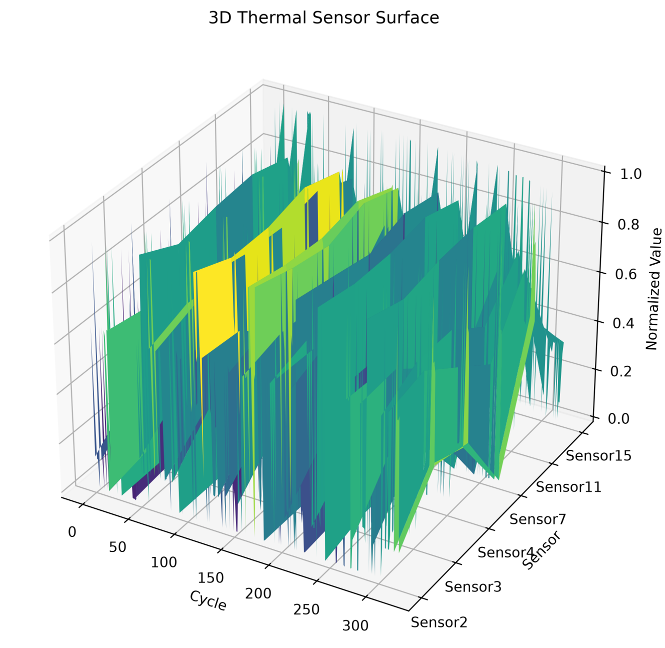

Fig. 15. Thermal surface visualization for NASA C-MAPSS FD004 dataset.

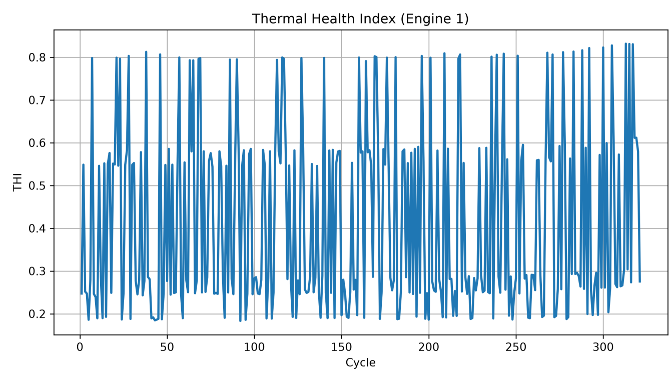

Fig. 16. Thermal health indicator analysis for FD004 dataset.

The dataset contains multivariate sensor measurements collected from simulated turbofan engines operating under different degradation conditions.

Two benchmark subsets are considered:

- FD001
- FD004

FD001 represents a relatively simple degradation scenario with a single operating condition and a single fault mode.

FD004 represents a more complex scenario containing multiple operating conditions and degradation patterns.

B. FD001 Analysis

The FD001 dataset is used as a baseline validation case for analysing thermal degradation behaviour.

The extracted thermal trends and health indicators provide information regarding system degradation progression under controlled operating conditions.

C. FD004 Analysis

FD004 provides a more challenging validation scenario due to the presence of multiple operating conditions and degradation modes.

The analysis of FD004 enables evaluation of QTherm-X under complex aerospace system behaviour.

D. FD001 and FD004 Comparison

A comparison between FD001 and FD004 datasets is presented in Table III.

The comparison demonstrates that FD004 provides a more challenging validation environment due to increased operational variability and degradation complexity.

The analysis highlights the potential of QTherm-X for future aerospace thermal monitoring and predictive maintenance applications.

TABLE III
COMPARISON OF NASA C-MAPSS FD001 AND FD004 DATASETS

Parameter	FD001	FD004
Operating Conditions	Single	Multiple
Fault Modes	Single	Multiple
Complexity Level	Low	High
Degradation Pattern	Simple	Complex
Validation Difficulty	Moderate	High
Application Scenario	Baseline Analysis	Advanced Prognostics

IX. CONCLUSION

This paper presented QTherm-X, an Explainable Quantum-Assisted Physics-Informed Neural Network framework for thermal physics applications.

The proposed framework integrates Variational Quantum Circuits with Physics-Informed Neural Networks to solve the one-dimensional heat equation while preserving governing physical constraints.

A benchmark comparison was performed between Classical PINN and Hybrid QAPINN architectures using different quantum circuit configurations.

The experimental results demonstrate that the Hybrid QAPINN successfully integrates quantum feature extraction with physics-informed learning and provides a foundation for quantum-assisted scientific machine learning.

The framework was further investigated for aerospace applications using NASA C-MAPSS FD001 and FD004 datasets, demonstrating its potential for thermal health analysis and predictive maintenance scenarios.

Future work will focus on:

- Scaling the framework to higher-dimensional partial differential equations.
- Evaluating larger quantum circuits on real quantum hardware.
- Integrating advanced thermal digital twin applications.
- Extending the framework toward real-time aerospace monitoring systems.

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